

Learning under Delayed Outcomes

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1 Introduction

In many settings, decision makers experience a delay before observing true outcomes. This can pose a challenge when real-time decision making is needed. Nevertheless, the decision maker may be able to observe a proxies for the outcome

1. In the case of disease forecasting, call volume of national medical hotline can be observed at time t but the true number of cases at time t cannot be observed.

How to learn from this information? We cannot expect the distribution over outcomes Y to be fixed over time.

2 Statistical Setting

Let $t = 1, 2, \dots, T$ denote time-steps and $i = 1, 2, \dots, n$ denote units. For each time-step t , there exists a joint distribution Q_t over a proxy outcome $Z \in \mathcal{Z}$ and a true outcome $Y \in \mathcal{Y}$. We observe Q_t for $t = 1, 2, \dots, m$. However, for time-steps $t > m$, we only observe $Q_{t,Y|Z \in \mathcal{A}}$, where $\mathcal{A} \subset \mathcal{Z}$.

We posit that the distribution over outcomes $Q_{t,Y}$ may evolve over time. Since $Q_{t,Y}$ evolves over time, we must also assume that the distribution of $Y | Z$ is fixed because there must exist some proxy variables z for which the conditional distribution of $Y | Z = z$ changes. Furthermore, we note that the distribution of $Z | Y$ may also change over time. At small time-steps t , the fraction of disease-related calls a call center receives may be high at all disease levels but at later time-steps t , the fraction of disease-related calls a call center receives may be much lower at all disease levels.

Nevertheless, a plausible assumption is that the relative probability of observing call fraction z given disease prevalence y and y' is fixed over time. Denote the relative probability of observing call fraction z as follows

$$w(y, y', z; t) = \frac{dQ_t(Z = z | Y = y)}{dQ_t(Z = z | Y = y')}.$$

Assumption 1. The function $w(y, y', z; t)$ is constant in t .

This assumption implies the following factorization $dQ_t(z | y) = w(y, z) \cdot \lambda(z, t)$. The following result shows how we can estimate w by pooling data from all time periods.

Since $dQ_t(z | y) = w(y, z) \cdot \lambda(z, t)$, we also have that $\frac{dQ_t(z|y)}{dQ_t(z)} = w(y, z) \cdot \tilde{\lambda}(z, t)$. We note that the left side corresponds to the density ratio between the joint distribution Q_t over (Z, Y) and the product of the marginal distribution $dQ_t(y) \cdot dQ_t(z)$.

$$\frac{dQ_t(z | y)}{dQ_t(z)} = \frac{dQ_t(z, y)}{dQ_{t,Z}(z) \cdot dQ_{t,Y}(y)}.$$

Thus, the density ratio between the joint distribution and the product of the marginal distributions can be expressed as

$$\frac{dQ_t(z, y)}{dQ_t(z) \cdot dQ_t(y)} = w(y, z) \cdot \tilde{\lambda}(z, t).$$

This yields a simple algorithm for estimating w . First, we construct a pooled data distribution F over (T, Y, Z, R) , where $R \sim \text{Bernoulli}(1/2)$, $T \sim \text{Uniform}([m])$, $F_{Y,Z|T=t,R=1} = Q_t$, and $F_{Y,Z|T=t,R=0} = Q_{t,Y} \cdot Q_{t,Z}$. Then we propose to solve

Theorem 1. *Suppose Assumption 1 holds. Define a loss function*

$$L(r, a, z) := \log(1 + \exp(r + a)) - z \cdot (r + a).$$

Then $w(y, z) := \exp(h^(y, z))$, where h^* is the solution to the following augmented convex minimization problem*

$$\{h^*(\cdot), \alpha^*(\cdot)\} \in \arg \min_{(h, \alpha) \in \mathcal{H} \times \mathcal{A}} \{\mathbb{E}_F [L(h(Z, Y), \alpha(Z, T), R)] : \mathbb{E}_F [\alpha(Z, T)] = 0\}.$$

3 References

3.1 Detecting and Correcting for Label Shift with Black Box Predictors

Lipton et al. [2018]

Key assumption is that $dP(z | y) = dQ(z | y)$ but $dP(y) = dQ(y)$ can change. They have training data $(X, Y) \sim P$, a model $\hat{Y} = f(X)$, and unlabeled target data $X \sim Q$. They are able to estimate $dQ(y)/dP(y)$ as follows:

$$\begin{aligned} \mathbb{P}_Q [\hat{Y} = i] &= \sum_j \mathbb{P}_Q [\hat{Y} = i | Y = j] \cdot \mathbb{P}_Q [Y = j] \\ &= \sum_j \mathbb{P}_P [\hat{Y} = i, Y = j] \cdot \frac{\mathbb{P}_Q [Y = j]}{\mathbb{P}_P [Y = j]} \\ &= \sum_j \mathbb{P}_P [\hat{Y} = i, Y = j] \cdot w(Y = j). \end{aligned}$$

They assume that the confusion matrix is invertible.

References

Zachary Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and correcting for label shift with black box predictors. In *International conference on machine learning*, pages 3122–3130. PMLR, 2018.